EXPERIMENT 12

Question: Write the Python program for the Tic Tac Toe game.

Aim: To write a Python program to implement the Tic-Tac-Toe game for two players and determine the winner or draw.

Algorithm:

1. Start the program.
2. Initialize a 3 × 3 game board with empty positions.
3. Display the board with position numbers from 1 to 9.
4. Allow the players X and O to enter their moves alternately.
5. Check whether the selected position is valid and empty.
6. After each move, check whether a player has completed a row, column, or diagonal.
7. If a winning condition is satisfied, display the winner.
8. If all positions are filled without a winner, declare a draw.
9. Stop the program.

Code:



Output:

```
>>> ================================ RESTART ================================
>>>
Positions: 1 to 9
  |   |
--+---+--
  |   |
--+---+--
  |   |
Enter your move (1-9): 1
Computer chooses: 5
Enter your move (1-9): 9
Computer chooses: 2
Enter your move (1-9):
>>> ================================ RESTART ================================
>>>
Positions: 1 to 9
  |   |
--+---+--
  |   |
--+---+--
  |   |
Enter your move (1-9): 1
Computer chooses: 5
Enter your move (1-9): 9
Computer chooses: 2
Enter your move (1-9): 3
Computer chooses: 6
Enter your move (1-9): 4
Computer chooses: 8
X | O | X
--+---+--
X | O | O
--+---+--
  | O | X
Computer wins!
>>>
```

Result: Thus, the Python program to implement the Tic-Tac-Toe game was executed successfully, and the winner or draw was determined based on the game conditions.